

Ramy Elsaraf

Software Developer | AI/ML Engineer
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PROFESSIONAL SUMMARY

Versatile software engineer and holder of two master's degrees with 3 years of experience in full-stack software development creating exceptionally impactful and optimized applications with robust user experiences. Solid proficiency in Python for developing machine learning and AI applications with practical experience in AI, machine learning, and deep learning frameworks including TensorFlow and PyTorch, LLM development with fine-tuning and prompt engineering, and deploying AI solutions to production environments on AWS, Google Cloud, and Azure.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript/TypeScript, Golang, C++, C#, Java, R, PHP, SQL
- **AI & Machine Learning:** TensorFlow, PyTorch, scikit-learn, Keras, Hugging Face, deep learning, NLP, computer vision, LLM development, fine-tuning, prompt engineering
- **Backend Development:** Python (Flask, Django, FastAPI), ASP.NET, Node.js, PHP Laravel, REST API
- **Frontend Development:** React.js, Next.js, Vue.js, HTML5, CSS3, JavaScript/TypeScript, AJAX
- **Cloud Platforms:** AWS, Microsoft Azure, Google Cloud Platform, cloud deployment and architecture
- **Databases:** SQL Server, MySQL, PostgreSQL, MongoDB, NoSQL, database design and optimization
- **DevOps & Tools:** Docker, Kubernetes, AKS, Git, Linux, Jira, Postman, Selenium, Firebase, CI/CD
- **Data & Analytics:** pandas, NumPy, SciPy, data preprocessing, feature engineering, data analysis
- **Methodologies:** Agile, DevOps, test automation, code review, project management

EDUCATION

Doctor of Philosophy (Ph.D.) in Computer Engineering (In Progress)

Memorial University of Newfoundland | St. John's, NL | 2026 - Present

- Focus: Artificial Intelligence and Machine Learning

Master of Applied Science (M.A.Sc.) in Computer Engineering

Memorial University of Newfoundland | St. John's, NL | September 2020 - June 2022

- GPA: 3.54/4.00 | Vice President of EAGE (European Association of Geoscientists and Engineers)
- Thesis: A Machine Learning Program for Detecting COVID-19 in Human Chest Image Scans (92% grade)
- Utilized TensorFlow, Deep Learning and deployed Machine Learning Program on Google Cloud

Master of Science (M.Sc.) in Microelectronic Systems

University of Liverpool | Liverpool, United Kingdom | September 2017 - December 2018

Bachelor of Science (B.Sc.) in Computer Science

The Open University | United Kingdom | June 2011 - September 2015

- First Class Honors | GPA: 3.53/4.00

PROFESSIONAL EXPERIENCE

Full Stack Software Developer

YSwipe | Calgary, Alberta | May 2024 - February 2026

- Developed YSwipe's web application UI with HTML, CSS, and JavaScript/TypeScript, ReactJS, NextJS ensuring responsive and seamless user experience

- Engineered backend of YSwipe's web application using Python FastAPI, incorporating features such as transcription and scoring mechanism for candidates
- Deployed web application using AWS
- Built innovative video platform with recording and rating functionalities tailored for candidate assessments and HR recruitment using AI
- Leveraged AI, Machine Learning, and Deep Learning technologies, along with AI Avatars, to enhance and evaluate candidates' performance
- Utilized TensorFlow, PyTorch, and gained familiarity with LLM development, fine-tuning, and prompt engineering
- Integrated third-party APIs and databases to support product functionality
- Participated in Agile development processes and code reviews

Software Developer

EARA Technologies Inc. | Calgary, Alberta | January 2023 - August 2023

- Contributed to broad range of technology projects including REST API's, web application components and endpoints using Python and Postman
- Crafted and ran Linux scripts, and optimized functions on users reducing runtime by 50%
- Worked both independently and with four other team members in DevOps environment using Kubernetes, Docker and AKS
- Utilized Azure Cloud Computing for deploying solutions in production
- Implemented 61 automated tests that improved test coverage from 40% to 100% using Python and Selenium
- Conducted troubleshooting, debugging, analysis, investigation, remediation, and communication to client and team

Full Stack Software and Web Developer

Gelismis Global Company | Istanbul, Turkey | November 2019 - November 2020

- Developed web applications' UIs and dashboards using ReactJS, VueJS, JavaScript/TypeScript, CSS, HTML, AJAX
- Developed and refactored REST APIs using Python, Laravel, ASP.NET and improved product retrieval by 100%
- Designed and refactored databases using MySQL and SQL Server and optimized execution time by 70%
- Used Jira for project management and implemented Agile software development
- Led team of four developers to create online automotive marketplace and other applications for various clients

PROFESSIONAL CERTIFICATES

Microsoft Programming in HTML5 with JavaScript and CSS3 | Microsoft Programming in C# and ASP.NET | Microsoft SQL Server | Microsoft Azure AI Fundamentals | Google Data Analytics Professional Certificate | IBM Data Analyst Professional Certificate | Cisco CCNA